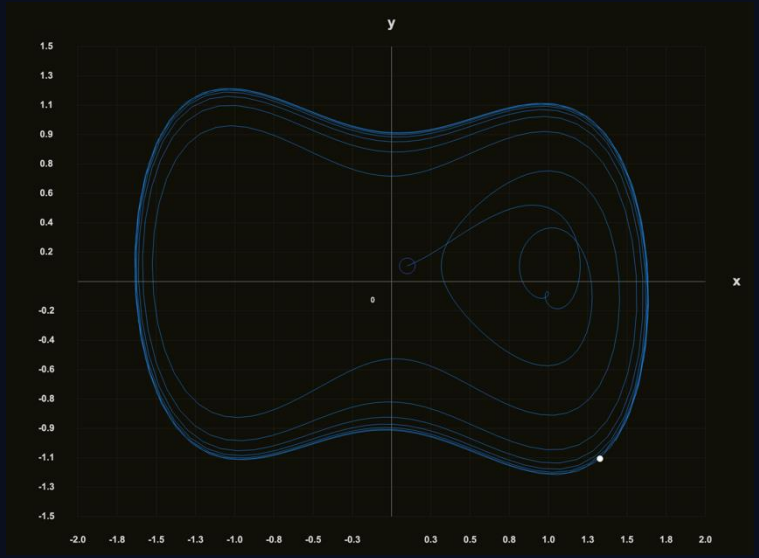
ACP2 Project Final Presentation

Project 06: Visualization of Set-valued Dynamical Systems



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Real Systems Are Never Perfectly Deterministic

Measurement noise

Sensors always have error — the state we observe is not the exact state.

Model uncertainty

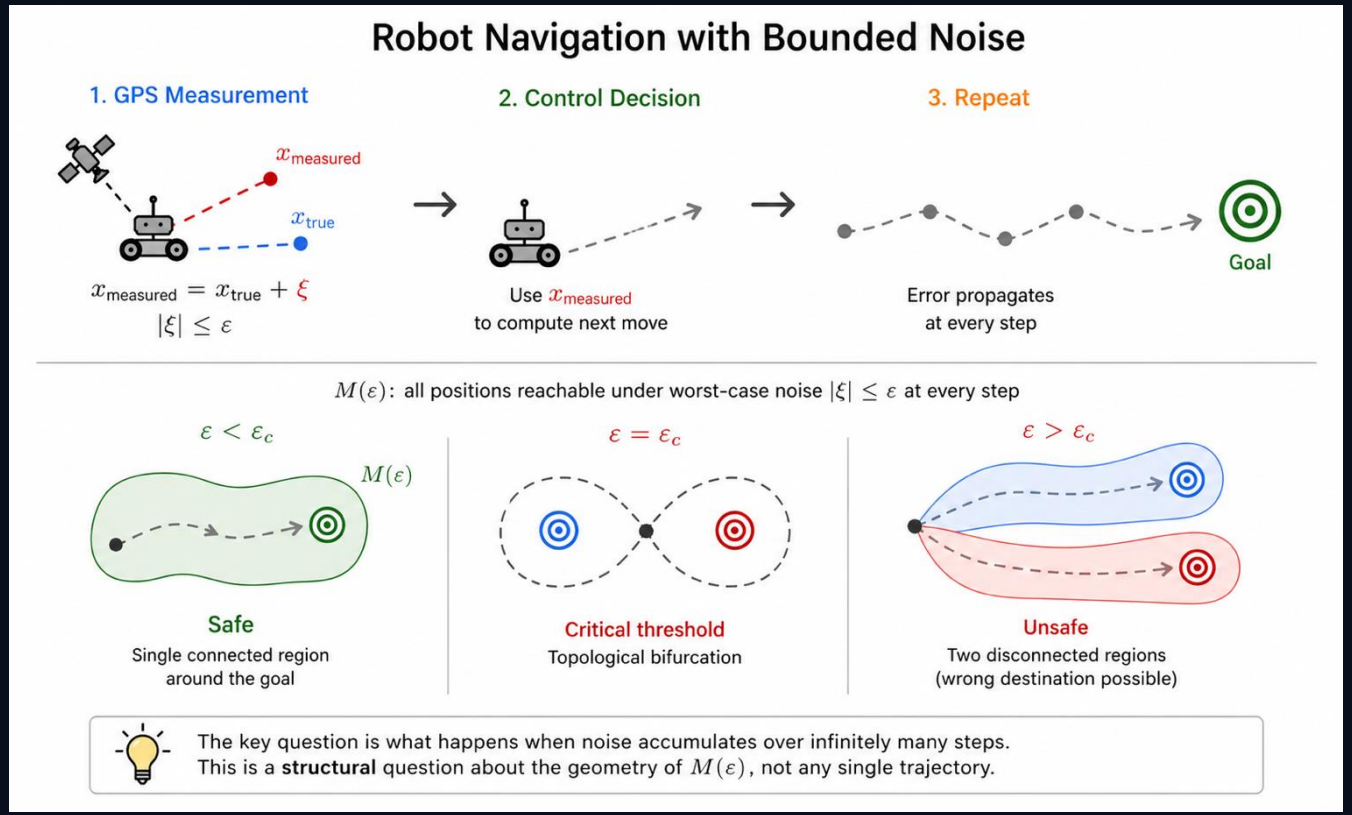
Physical models are approximations. Parameters drift over time.

Impact on stability

Where can the system end up in the long run, under worst-case bounded perturbations?

Classical deterministic analysis breaks down when noise is present.
We need a framework that treats noise as a structural property.

A real example: Robot Navigation



Set-Valued Dynamical Systems

Deterministic map

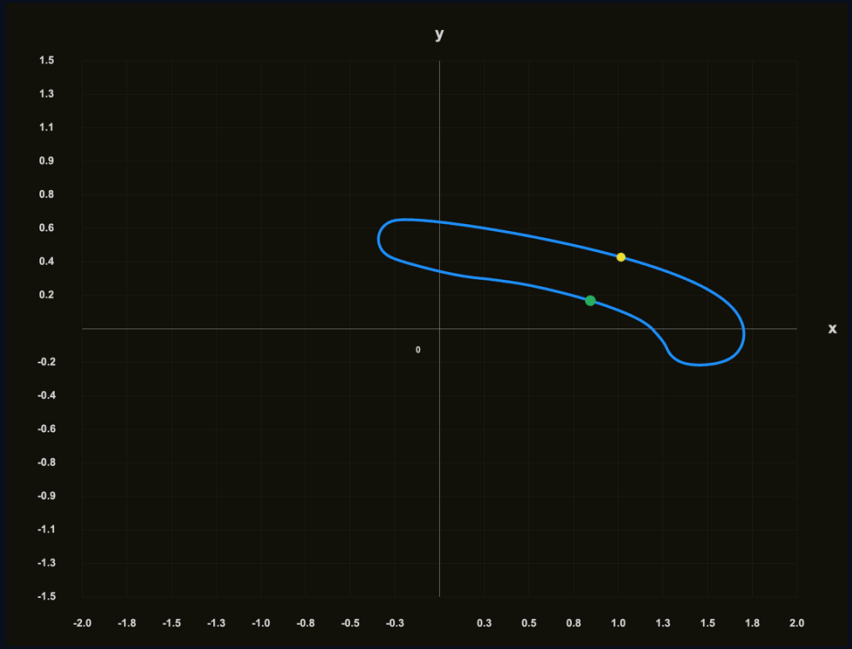
$$x \to f(x)$$

a single point maps to a single point

Set-valued map : $\epsilon > 0$

$$F_\epsilon(x) = B_\epsilon(f(x))$$

Every image point is replaced by a ball of radius ε — the maximum noise the system can experience.



Hénon map · a=0.4, b=0.3, ε=0.1

The Minimal Invariant Set M(ε)

The smallest compact set that maps into itself : $F_\epsilon(M) = M$

Reachability Envelope

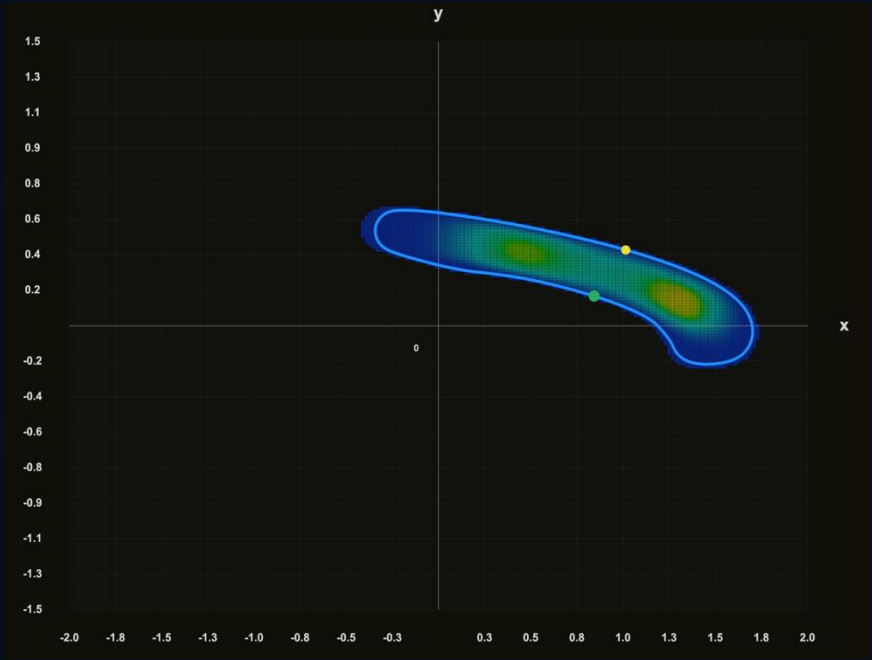
The exact outer boundary of every state the system can ever visit under noise ≤ ε.
Starting inside M → staying inside M.

Stationary Measure

M(ε) is the support of the stationary probability measure — where the system spends its time in the long run.

Structural Stability Radius

The noise level ε at which M(ε) changes topology tells us how resilient the attractor is.



MIS boundary + Ulam stationary measure

How Do We Compute ∂M(ε)?

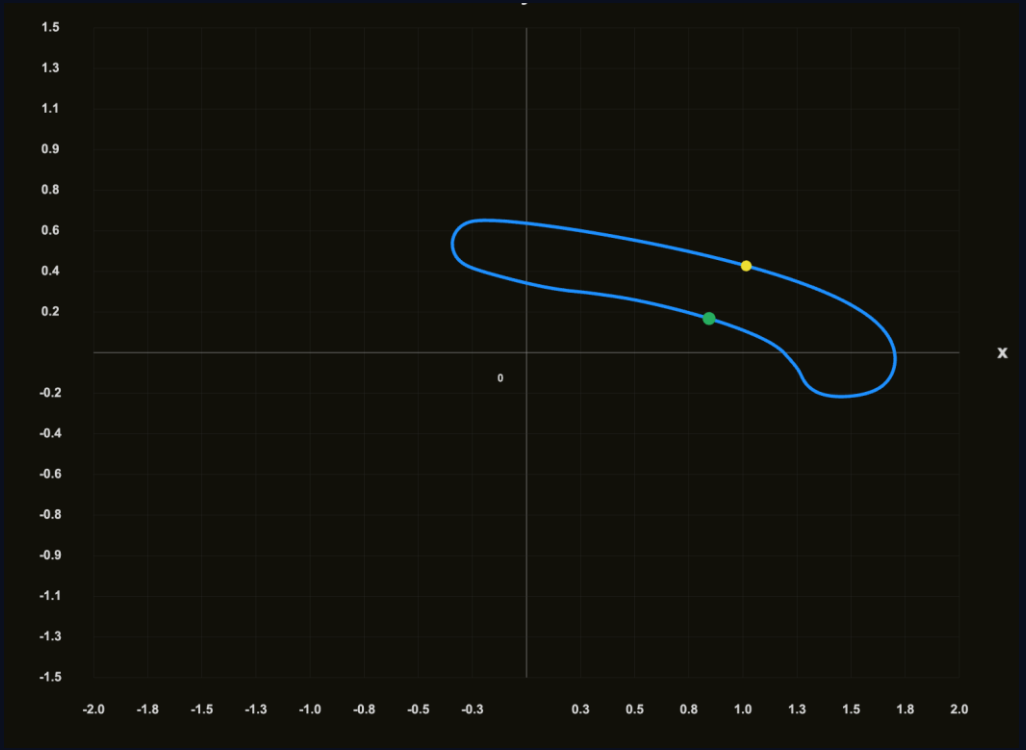
Key insight: the boundary map E on $\mathbb{R}^2 \times S^1$ is deterministic.

1. Augment state space to(x,n) where n is the unit outward normal

2. Define boundary map$E(x, n) = \left(f(x) + \varepsilon \cdot \frac{J^{-T}n}{\|J^{-T}n\|}, \frac{J^{-T}n}{\|J^{-T}n\|}\right)$ on $\mathbb{R}^2 \times S^1$.

3. Find periodic orbits of E using the Davidchack–Lai stabilized Newton solver

4. Grow unstable manifolds from those orbits– they trace $\partial M(\varepsilon)$ exactly.



Unstable manifolds tracing ∂M(ε)

The Gap: Existing Tools vs. BIST

GAIO (MATLAB)

✓ Complete algorithm suite

✓ Proven in research

✗ Requires MATLAB install

✗ Steep learning curve

✗ No interactivity

Python Scripts

✓ Accessible language

✓ Flexible

✗ No GUI

✗ Static single-use scripts

✗ Manual parameter changes

BIST

✓ Zero install — runs in browser

✓ Real-time interactive

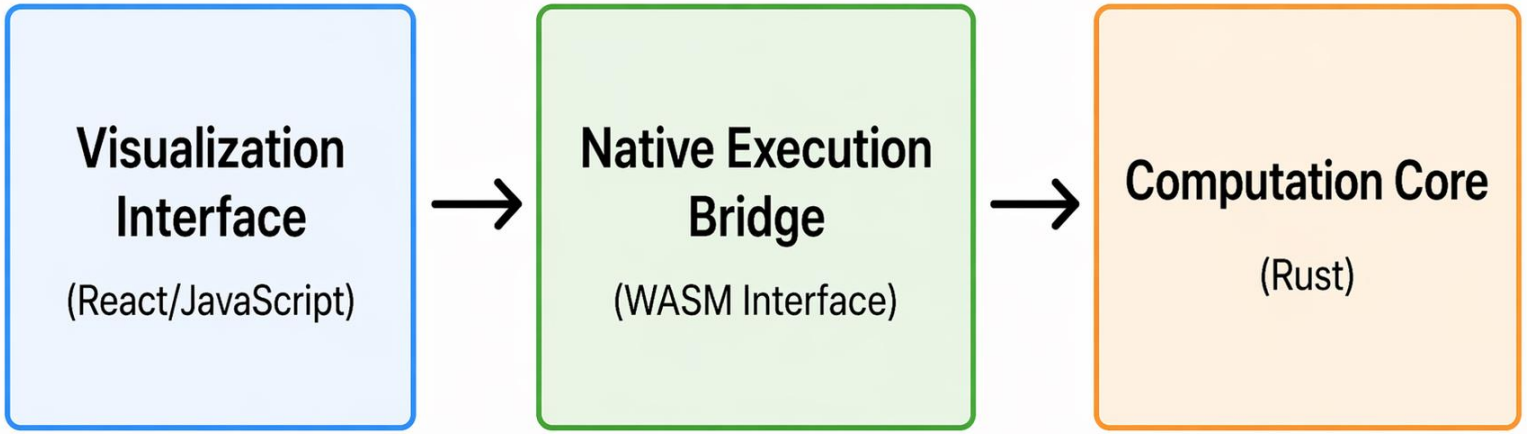
✓ Custom equations

✓ Both discrete & continuous

✓ Topological bifurcation analysis with MIS and stationary measures

BIST bridges the gap: accessible to anyone with a web browser.

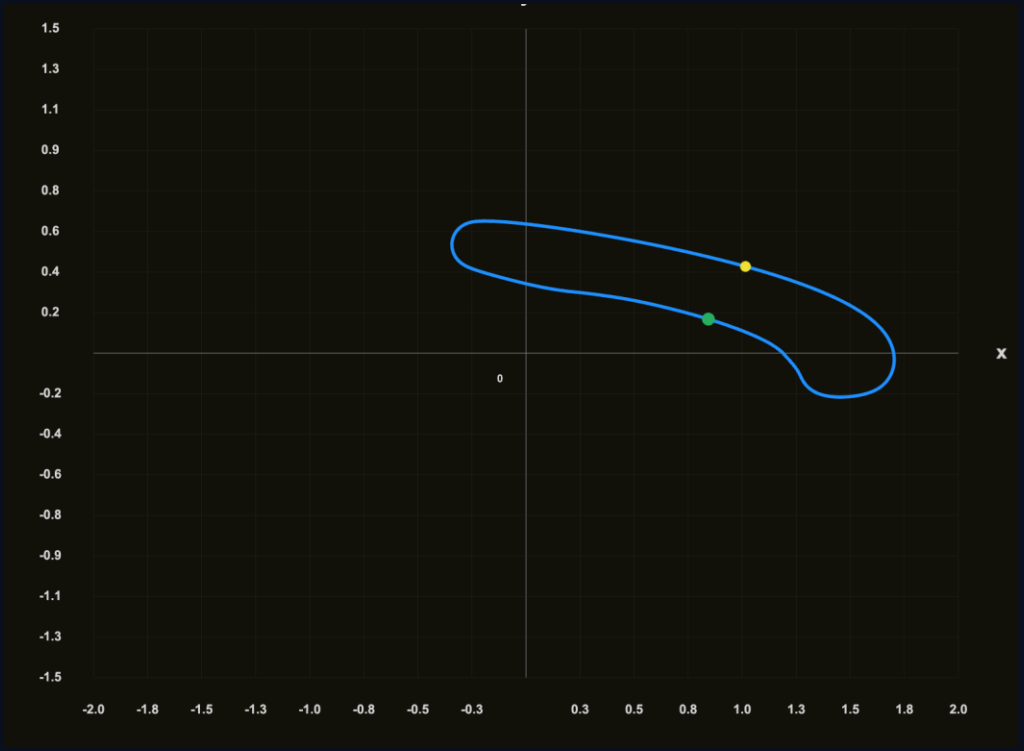
How BIST Is Built



Demo 1: Minimal Invariant Set

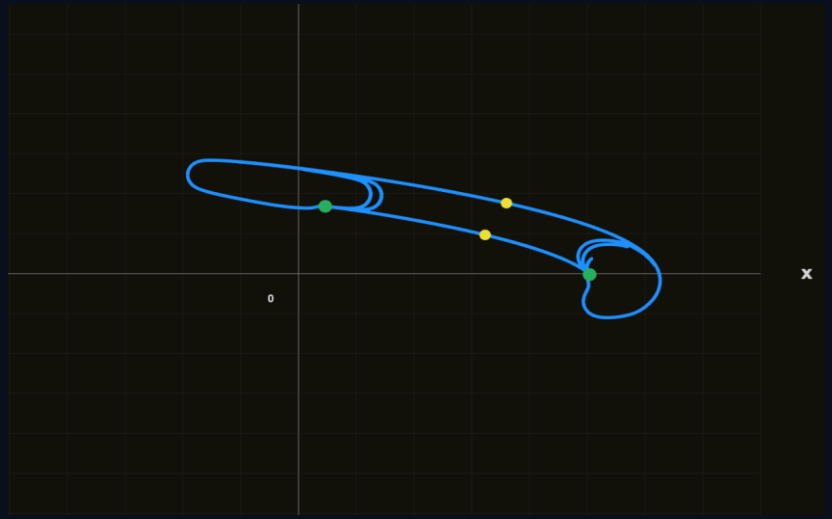
Hénon Map · a = 0.4, b = 0.3, ε = 0.0625

The blue closed curve is ∂M(ε) — the exact boundary of the minimal invariant set.
The system cannot escape this region under noise amplitude ≤ ε.



Demo 2: Topological Bifurcation

b = 0.3, ε = 0.0625 · increasing a past a critical value splits M(ε) into two disjoint regions



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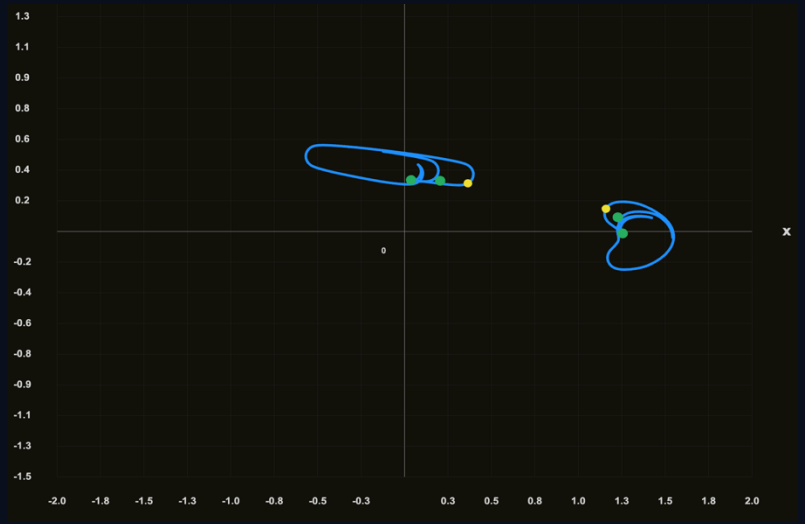
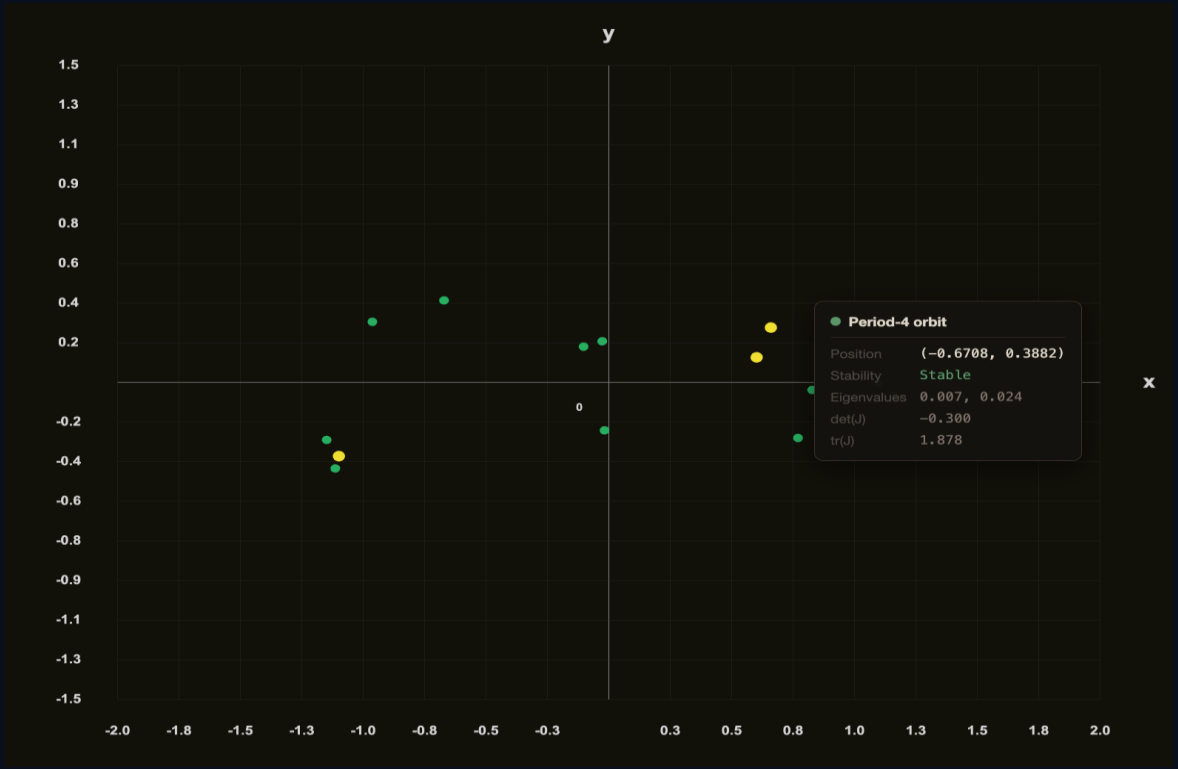


Before · a = 0.55 → single connected region

After a = 0.60 → splits into TWO disjoint regions

Demo 3: Periodic Orbit Detection



Periodic Orbit finder with Newton's method and Davidchack–Lai stabilized solver

finds all periodic orbits in the system with a custom grid search resolution

Demo 4: Stationary Measure via Ulam's Method

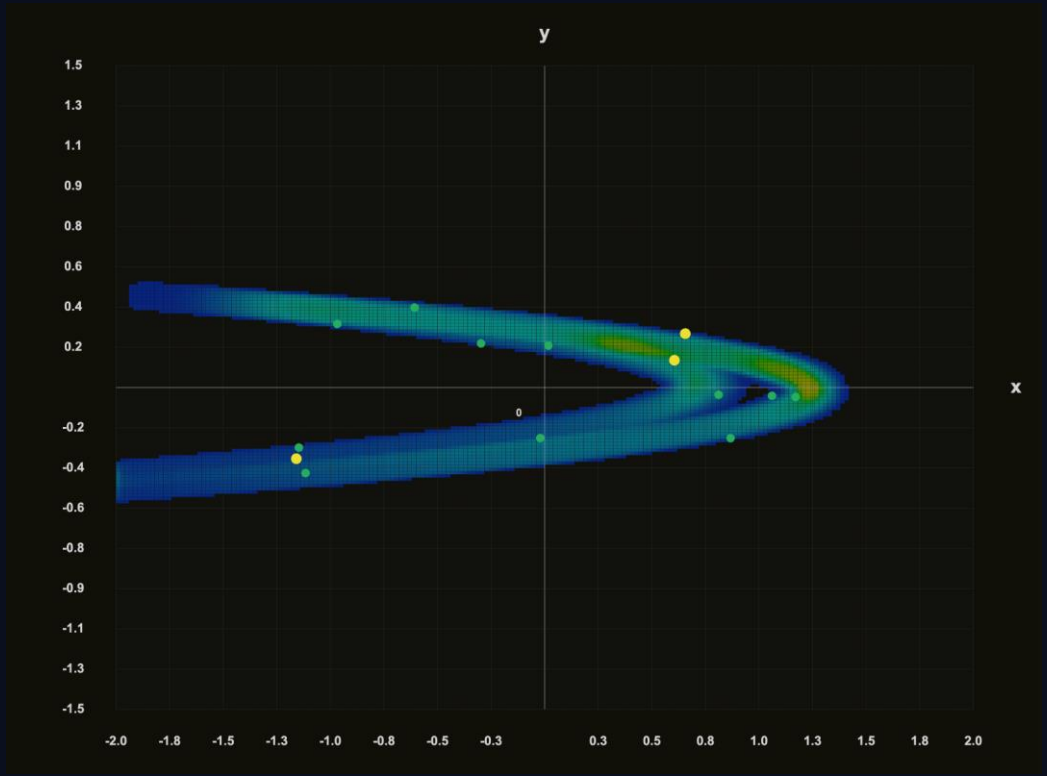
What Ulam's method does:

1. Partition phase space into boxes

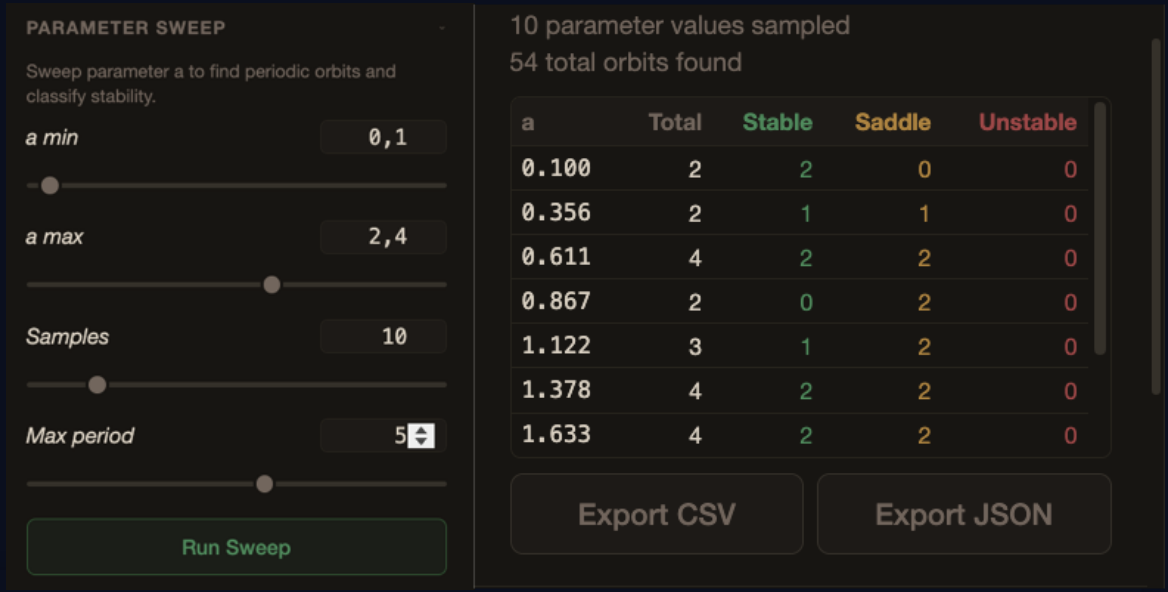
2. Build a transition matrix — how likely is it to jump from box i to box j?

3. Find the stationary distribution (left eigenvector)

4. Render as a heatmap → shows where the system spends its time



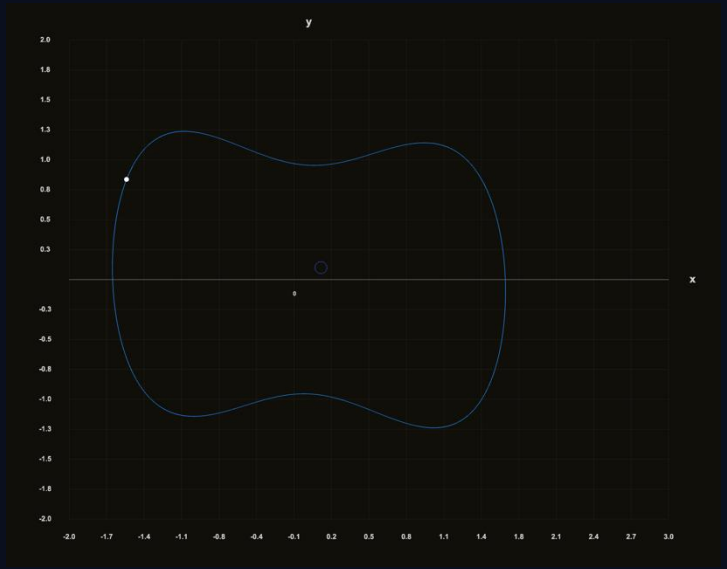
Demo 5: Automated Parameter Sweep



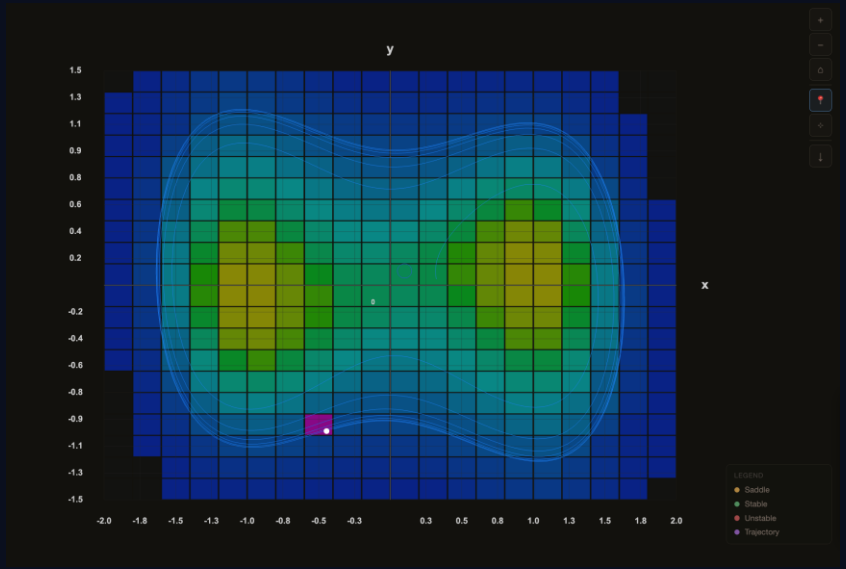
a	Total	Stable	Saddle	Unstable
0.100	2	2	0	0
0.356	2	1	1	0
0.611	4	2	2	0
0.867	2	0	2	0
1.122	3	1	2	0
1.378	4	2	2	0
1.633	4	2	2	0

Sweep any parameter (a, b, ε) and find out the periodic orbits. Results export to CSV, JSON for further analysis.

Demo 6: Continuous Systems — Duffing Oscillator



MIS boundary - Duffing ODE

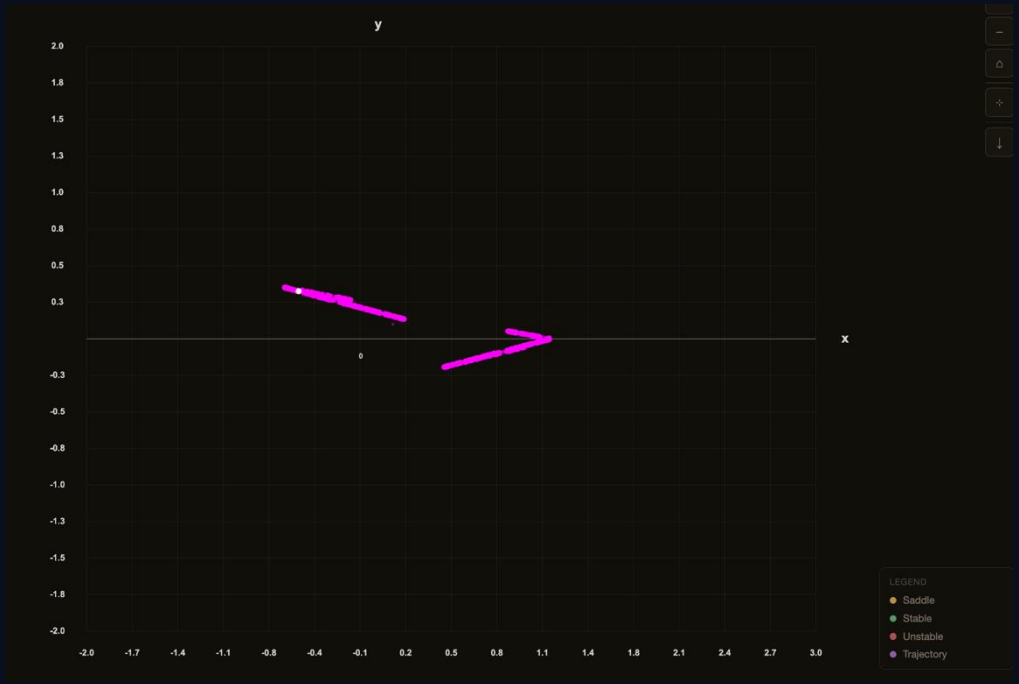


Ulam stationary measure — Duffing ODE

Demo 7: Custom Equations — Lozi Map

Support some custom 2D maps,
e.g.: the Lozi map

$$f_x = 1 - 1.4 * |x| + y$$
$$f_y = 0.3 * x$$

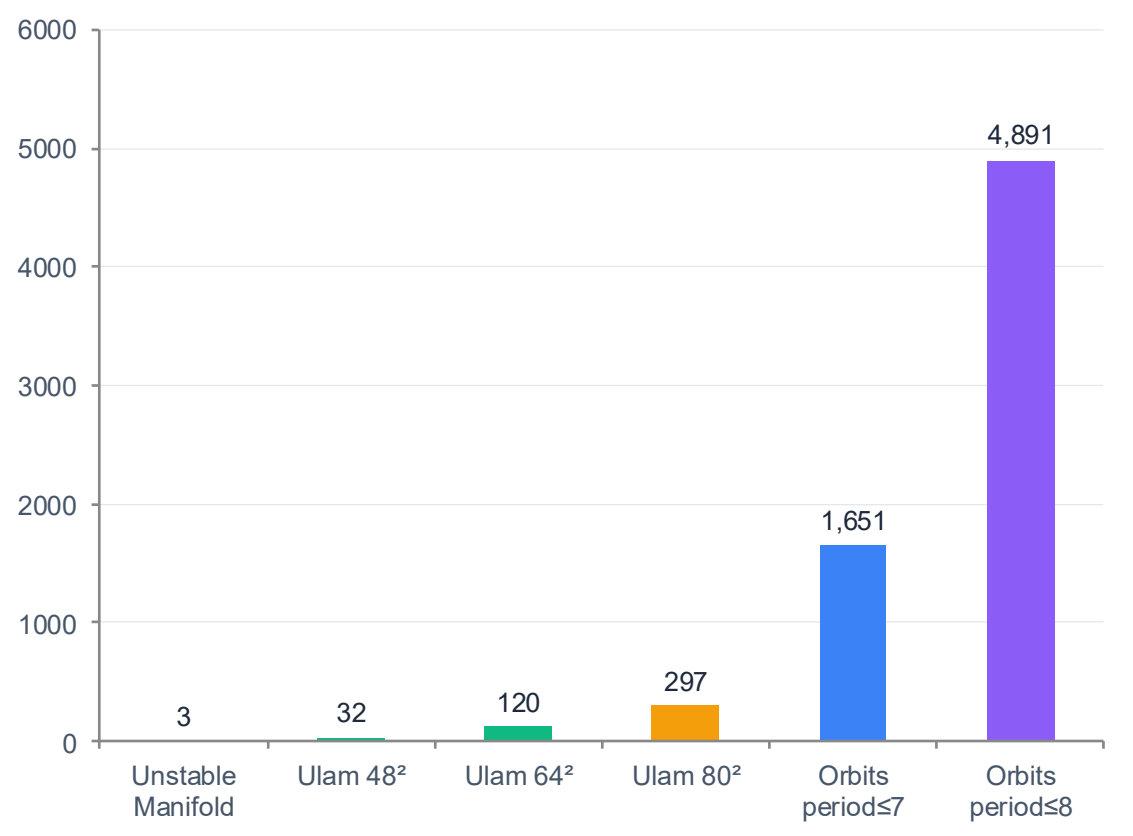


Evaluation

92	72	0	164
Rust unit tests	Frontend tests	Failures	Total · all pass

- ✓ At $\varepsilon=0$, boundary map reduces exactly to $f(x)$ — verified analytically
- ✓ Single connected MIS at $a=0.4$, $b=0.6$, $\varepsilon=0.065$ — matches Tey (2023)
- ✓ Topological bifurcation at $a=0.6$, $b=0.3$, $\varepsilon=0.0625$ — matches Lamb, Rasmussen & Tey (2025)
- ✓ Lozi map via user-defined input matches analytic attractor shape
- ✓ User-defined Hénon matches hard-coded implementation to 10^{-10} residual

Performance · Apple M1 Pro



Manifold: < 5 ms

Sequential forward iteration with early stopping — negligible cost.

Ulam: $O(n^2)$

Doubling grid size from 48→80 ≈ 9× slower — matches theory.

Orbit search dominates

99% of run time — once per session, not on every render.

Conclusions & Future Work

What we built

→ Browser-native tool, zero install required

→ Boundary map + unstable manifold computation

→ Ulam stationary measure with heatmap display

→ Davidchack–Lai periodic orbit solver

→ Automated bifurcation sweep with export

→ Discrete maps + continuous ODEs (RK4)

→ Custom symbolic equation pipeline

Future directions

- Non-uniform noise $\varepsilon(x)$
- Visualize arbitrary invariant sets, not just MIS
- Server-side backend for large-scale research
- WebGPU for parallel Ulam matrix assembly
- 3D phase space visualization

BIST makes the mathematics of bounded noise accessible.
From a rough Python prototype to a browser-native research tool — thanks to continuous feedback and contribution from Dr. Kalle Timperi and researchers at Imperial College London.

namvdo.github.io/set-valued-viz